Relational σ-Additive Probability on Orthomodular Lattices: A Working Problem-Set

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Abstract

A self-contained kit for attacking the descent-axis open problem under the *Reading-1* (relational = no hidden realisation) framing. Everything needed to sit down with pen and paper is defined inline: orthomodular lattices and the orthogonal/meet-zero gap; states, dispersion-free states, concreteness; contextuality and the Fine equivalence; the centre and the Stone/Loomis–Sikorski correspondence; and the construction $\prod_n \mathrm{MO}_2$. The open target is stated formally (§7), reduced to a single existence question, and broken into concrete sub-tasks (§8) carrying the constraints established so far (the Wright bound; the off-centre requirement from the CE-routing analysis). Companion narrative: `oml_onboarding.{tex,md}`, `reading1_prize_reduction.md`, `sigma_essential_nonemptiness_finding.md`. This document is presentation, not proof: open questions are flagged as open; included proofs are short and self-verifying or cited.

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Why this matters

The question behind the question. Classical probability is built on a sample space: a set Ω of definite states, with events as its subsets. A measure is then an average over those states. The programme this problem-set serves asks whether probability can be had *without* that scaffolding — *relationally*, from the entailment order of propositions alone, with no underlying space of hidden definite states to descend from. In the Boolean (classical) case the answer is yes and well understood (this is the content of the CE / Paper-I results: coherence forces a unique σ-additive measure, no Ω assumed). The open frontier is the *non-Boolean* case, where some propositions are *incompatible* — not jointly decidable — so the event algebra is an orthomodular lattice rather than a Boolean one. There, classical probability has no room for the incompatibility, and whether an honest (σ-additive) relational probability theory exists at all is unsettled.

Why "contextual" is the precise form of "relational." A measure that is secretly an average over hidden definite states is *not* relational — it is classical probability in disguise. By Fine's theorem (Thm. 3.2) "average over hidden states" is exactly *non-contextual*. So the relational prize is a *contextual* σ-additive state: probability with no global hidden joint, genuinely on the non-distributive lattice. This is the measure-theoretic analogue of *localic* (point-free) topology — spaces recovered from their lattices of opens — but in the one regime localic measure theory does not reach: the *non-distributive* one.

What an answer decides. A *witness* (§8, Exit A) would be the first genuine relational σ-additive probability on a non-Boolean lattice — a new object, and a positive resolution of a two-year programme thread. An *impossibility proof* (Exit B) would be just as decisive in the other direction: it would show that the relational content the programme seeks is *operationally invisible* — present finitely, but unable to survive the passage to the countable level — turning a philosophical worry into a theorem. The finite case is already settled (Wright's pentagon, §6); the entire live question is whether the phenomenon can be *σ-essential*, i.e. forced by the infinite structure and not visible in any finite piece. That "finite coherence vs. the countable commitment" shape is the same one at the heart of CE / Paper I — which is why this problem, though it wears quantum-logic clothing, rejoins the programme's origin rather than departing from it.

1 Orthomodular lattices and the orthogonal/meet-zero gap

Definition 1.1 (Orthomodular lattice)**.** An *orthomodular lattice* (OML) is a bounded lattice $(L, \wedge, \vee, 0, 1)$ with an order-reversing involution $a \mapsto a^\perp$ such that

$$a \vee a^\perp = 1, \qquad a \wedge a^\perp = 0, \qquad (a^\perp)^\perp = a,$$

and satisfying the *orthomodular law*:

$$a \leq b \implies b = a \vee (b \wedge a^\perp).$$

Write $a \perp b$ ("*a orthogonal b*") for $a \leq b^\perp$. L is *Boolean* iff it is distributive.

Definition 1.2 (The two relations, and the gap)**.** For $a, b \in L$:

$$a \perp b \;:\!\iff\; a \leq b^\perp \qquad \text{(orthogonal)}, \qquad\qquad a \wedge b = 0 \qquad \text{(meet-zero)}.$$

Always $a \perp b \Rightarrow a \wedge b = 0$. The converse *fails* in a non-Boolean OML; the failure is the source of everything here.

Example 1.3 (MO_2, the minimal gap)**.** $\mathrm{MO}_2 = \{0, 1, a, a^\perp, b, b^\perp\}$ (the "Chinese lantern"): four atoms, each pair of distinct atoms joining to 1 and meeting to 0. So $a \wedge b = 0$ but $a \not\leq b^\perp$ (as $b^\perp \in \{b^\perp\}$ and $a \neq b^\perp$), i.e. $a \wedge b = 0$ yet $a \not\perp b$. MO_2 is the smallest non-Boolean OML and the unit of every construction below.

Definition 1.4 (Completeness notions)**.** L is *σ-complete* if every countable family $\{a_n\} \subseteq L$ has a join $\bigvee_n a_n \in L$. L is *σ-orthocomplete* if this holds for every countable *pairwise-orthogonal* family. (σ-complete $\Rightarrow$ σ-orthocomplete; the converse fails on non-Boolean OMLs.)

Definition 1.5 (Centre)**.** $z \in L$ is *central* if it is compatible with every element and $L \cong [0, z] \times [0, z^\perp]$ as OMLs. The *centre* $Z(L) = \{\text{central elements}\}$ is a Boolean subalgebra; L is *irreducible* if $Z(L) = \{0, 1\}$.

2 States, dispersion-free states, concreteness

Definition 2.1 (State)**.** A *state* on L is $s\colon L \to [0,1]$ with $s(1) = 1$ and finite additivity on orthogonal pairs: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$. It is *σ-additive* if $s(\bigvee_n a_n) = \sum_n s(a_n)$ for countable pairwise-orthogonal $\{a_n\}$ (join taken in L). A state is *dispersion-free* (2-valued) if $s(L) \subseteq \{0,1\}$.

Definition 2.2 (Concrete logic)**.** A set S of states is *order-determining* if $\big(\forall s \in S : s(a) \leq s(b)\big) \Rightarrow a \leq b$. L is *concrete* (equivalently *set-representable*) iff it admits an order-determining set of *dispersion-free* states (Gudder [3]). Equivalently, L embeds into some $(\mathcal{P}(X), \subseteq, X \setminus (-))$ preserving existing orthogonal joins.

Remark 2.3 (Separating vs. spanning — the load-bearing distinction)*.* Concreteness asks the dispersion-free states to *separate* the order. It does *not* ask them to *span* the state space (i.e. that every state be a mixture of them). The whole problem lives in this gap; keep it in view.

Fact 2.4 (MO_2 has no dispersion-free homomorphism, but is concrete)**.** MO_2 *has exactly four order-determining dispersion-free* states *(each sending one atom-pair to* 0 *and the other to* 1*), so* MO_2 *is concrete. None is a lattice* homomorphism*: for the state* s *with* $s(a) = s(b) = 0$ *one has* $s(a \vee b) = s(1) = 1 \neq 0 = \max(s(a), s(b))$ *(equivalently* $1 > 0 = s(a) + s(b)$*, so* s *is not subadditive).* Concrete uses states, not homomorphisms.

3 Contextuality and the Fine equivalence

The Reading-1 reading of "relational" — the measure carries *no hidden realisation* — is made precise by contextuality. Fix a concrete OML L with order-determining dispersion-free set S_{df}.

Definition 3.1 (Contextual state)**.** A σ-additive state w on L is *non-contextual* if it lies in the closed convex hull $\overline{\mathrm{conv}}(S_{\mathrm{df}})$ of the dispersion-free states (in the topology of pointwise convergence on L); otherwise *contextual.*

Theorem 3.2 (Fine, in lattice form)**.** *For a concrete L, the following are equivalent for a σ-additive state w:*

(i) *w is non-contextual ($w \in \overline{\mathrm{conv}}(S_{\mathrm{df}})$);*

(ii) *$w = \int_{S_{\mathrm{df}}} s\, d\mu(s)$ for a probability measure μ on S_{df} (a* global hidden joint*);*

(iii) *the context-wise distributions w induces on the Boolean blocks of L admit a single global joint distribution reproducing all of them.*

A dispersion-free state is *a deterministic global section, so a global joint and a hull-decomposition are the same object (Fine [2]; Abramsky–Brandenburger [1]).*

Remark 3.3 (Three point-spaces — do not conflate)*.* The recurring trap. Distinguish:

(A) the *external* points (e.g. Hilbert rays of H for $L(H)$): an auxiliary space the measure could be reduced to (Gleason's ρ in $s = \mathrm{tr}(\rho\,\cdot)$). Reading-1 *forbids* reduction to these.

(B) the *canonical dual* points $P(A) = \{\uparrow p : p \in L\}$ (principal filters — one per *element*): built from L, hence *permitted.*

(C) the *dispersion-free states* S_{df} (one per consistent global *valuation*): the hidden-variable points of Def. 3.1.

"Relational = contextual" is a condition on (C), *not* on (B): a witness is point-*rich* on $P(A)$ yet has no global S_{df}-joint. (A), (B), (C) are genuinely different objects.

The open target. Relational σ-additive probability = a contextual σ-additive state on a concrete OML. (The formal open problem is §7.)

4 The Boolean engine, and why it does not transfer

Theorem 4.1 (Stone / Loomis–Sikorski). *For a Boolean algebra B with Stone space* $\mathrm{St}(B)$*: B is σ-complete iff* $\mathrm{St}(B)$ *is* basically disconnected *(the closure of every cozero set is open); and a finitely-additive state on B is σ-additive iff, viewed as a measure on* $\mathrm{St}(B)$*, it concentrates on the* principal *(isolated/atomic) part. In particular a σ-additive state on $\mathcal{P}(\mathbb{N})$ is atomic: $w = \sum_n w(\{n\})\delta_n$.*

Theorem 4.2 (Pták–Pulmannová [4]). *An OML is Boolean iff it carries a unital set of* subadditive *states. Hence the Boolean-forcing property is* subadditivity, *not σ-additivity: σ-additivity alone does not collapse a non-Boolean OML, so non-Boolean σ-additive states genuinely exist.*

Remark 4.3 (The Boolean kill-mechanism, and its Boolean-only hinge). On a Boolean algebra the chain "σ-additive $\Rightarrow$ concentrates on principal ultrafilters $\Rightarrow$ non-contextual" is forced, because there

$$\text{principal ultrafilter} = \text{atom} = \text{dispersion-free state}.$$

This triple identity is *Boolean-only.* On an OML with centre $Z(L)$: w restricted to $Z(L)$ is a σ-additive Boolean measure, so (by Thm. 4.1) it concentrates on the *central atoms* — but a central atom z subtends a (possibly non-Boolean) block $[0, z]$, and a state with $w(z) = 1$ can be *contextual within* $[0, z]$. So

$$\underbrace{\sigma\text{-add} \Rightarrow \text{concentrate on central atoms}}_{\text{transfers}}, \qquad \underbrace{\text{concentrate} \Rightarrow \text{non-contextual}}_{\text{Boolean-only; FAILS}}.$$

For *irreducible* L ($Z(L) = \{0, 1\}$) the centre argument says only $w(1) = 1$ — vacuous. *The Boolean kill-mechanism evaporates exactly off-centre, where a witness would live.* (Routing verdict: `sigma_essential_nonemptiness_finding.md`.)

Remark 4.4 (Why no faithful set representation — the universal floor). A faithful σ-tribe / Loomis–Sikorski representation realises L as a field of sets, preserving $\wedge, \vee$ as $\cap, \cup$; its image is a sublattice of a Boolean algebra, hence distributive, forcing L Boolean. So *no* non-Boolean OML admits one. A relational measure can therefore never arise from a faithful descent to points — the target is necessarily "point-free" in the sense of Rmk. 3.3(A), *not* of (B).

5 The construction $\prod_n \mathrm{MO}_2$, and what it settles

Definition 5.1 ($\prod_n \mathrm{MO}_2$). The full direct product of countably many copies of MO_2, coordinatewise operations. It is concrete (coordinate states separate), σ-complete (product of finite complete lattices), non-Boolean, infinite. Its centre contains $\prod_n\{0, 1\} \cong \mathcal{P}(\mathbb{N})$.

Proposition 5.2 (The interleaving property ($\bigstar$) is cheap). *There exist p and an infinite orthogonal family $\{a_n\}$ in $\prod_n \mathrm{MO}_2$ with $p \wedge a_n = 0$ and $p \not\leq a_n$ for all n. (Take $a_n = a$ in coordinate n, 0 elsewhere; $b_n = b$ in coordinate n, 0 elsewhere; $p = \bigvee_n b_n = (b, b, b, \dots)$, a genuine orthogonal join. Coordinate n: $a \wedge b = 0$ but $b \not\leq a^\perp$.) So "($\bigstar$) + concrete + σ-complete + non-Boolean" is satisfied by a one-line object: it is* not *the discriminator.*

Proposition 5.3 ($\prod_n \mathrm{MO}_2$ is not a witness — contextuality is only finitely additive). *Every σ-additive state on $\prod_n \mathrm{MO}_2$ is non-contextual (in $\overline{\mathrm{conv}}(S_{\mathrm{df}})$). Sketch: a contextual state would have to act diffusely on the central $\mathcal{P}(\mathbb{N})$ (a free-ultrafilter/Banach-limit type state); but by Thm. 4.1 a σ-additive state on $\mathcal{P}(\mathbb{N})$ is atomic, hence concentrates on coordinates, hence is a countable mixture of coordinate dispersion-free states. The contextual states exist only in the* finitely additive *layer; σ-additivity kills them. (This is the CE / Paper-I compactness phenomenon, and the reason $\prod_n \mathrm{MO}_2$ fails: all its infinitary structure is* central.*)*

6 The bound: Wright's pentagon

Theorem 6.1 (Wright [5]). *There is a* finite*,* concrete *OML — the pentagon (pentagram) logic, a Greechie loop of length five (hence a lattice) — with an order-determining family of dispersion-free states, carrying a state* not *in their convex hull, i.e. a* contextual *state.*

Remark 6.2 (What Wright settles, and what it leaves). A finite OML is vacuously σ-complete and its states vacuously σ-additive. So "concrete σ-complete non-Boolean OML with a σ-additive contextual state" is *already inhabited* by Wright — and the general spanning theorem "every state is a dispersion-free mixture" is *false.* What Wright does *not* give: contextuality requiring the *countable* structure. Its contextuality is witnessed by a finite sub-OML (itself). That single missing qualifier — **σ-essential** — is the entire open content.

7 The open target

> *The open target.* (**σ-essential contextuality).** Does there exist a concrete, σ-complete, non-Boolean OML L and a σ-additive state w on L such that:
> (1) w is *contextual* — $w \notin \overline{\mathrm{conv}}(S_{\mathrm{df}})$ (no global hidden joint over the dispersion-free states); and
> (2) the contextuality is σ*-essential* — the restriction of w to *every finite sub-OML* of L *is* non-contextual (so the obstruction lives only in the countable structure)?
> Equivalently: a relational σ-additive probability whose relationality is not already present finitely. **Or:** prove no such (L, w) exists.

Remark 7.1 (Status). **Well-posed, principled, uninhabited.** No witness ($\prod_n \mathrm{MO}_2$ fails by Prop. 5.3; Wright is finite, Rmk. 6.2); no impossibility proof. The CE-routing analysis (Rmk. 4.3) returned *GAP*: CE neither forbids nor produces a witness, but localises both exits (§8).

8 Sub-tasks — the pen-and-paper frontier

Two exits; each is genuine mathematics, not orientation. The constraints below are established (Rmk. 4.3, Prop. 5.3) and prune the search.

Exit A — build a σ-essential witness

Sub-task 1 (The off-centre constraint). Show (or use) that a witness must place its infinite non-Boolean structure *off-centre*: L irreducible, or at least with the relevant infinite blocks non-central. *Why:* if all infinitary structure is central, Stone over the centre (Thm. 4.1) forces atomicity and kills contextuality σ-additively (Prop. 5.3). *Excludes products of finite blocks a priori.*

Sub-task 2 (Pasting / countable colimit of Wright blocks). Construct an infinite concrete σ-complete OML as a countable paste/colimit of Wright-type (pentagon) blocks such that:
(a) each finite sub-OML is non-contextual (dispersion-free states span its state space);
(b) the centre stays trivial (or the relevant blocks non-central) — Task 1;
(c) the countable join forces a contextual σ-additive state (a Kochen–Specker-type obstruction appearing only in the limit).

Concretely: can the Greechie/pasting machinery that yields the finite pentagon be iterated to a σ-orthocomplete limit while keeping the centre trivial and concreteness intact? CE raises no objection (Rmk. 4.3); this is the sanctioned shape. *One such witness settles the descent axis positively.*

Exit B — prove no witness exists

Sub-task 3 (A new finite-witnessing theorem)**.** Prove: every σ-additive contextual state on a concrete σ-complete OML is *already* contextual on some finite sub-OML (equivalently, σ-essential contextuality is impossible). **Constraint:** this *cannot* reuse the $\prod_n \mathrm{MO}_2$/Stone-over-centre mechanism — it is vacuous off-centre (Rmk. 4.3). A genuinely different argument is required. *Lever:* the Pták–Pulmannová subadditivity boundary (Thm. 4.2); the missing bridge is subadditivity $\leftrightarrow$ contextuality at the σ level. *Such a theorem reinstates the empiricist-underdetermination conclusion as a theorem* (`direction2_gate_finding.md`).

Remark 8.1 (Discipline). Neither exit is decidable by a quick computation; both are Phase-4 mathematics. The recurring failure mode in this material is the clean "X, therefore Y" step that needs a qualifier (the Boolean triple identity of Rmk. 4.3 is the paradigm). Verify transfers; do not assume them. A candidate witness is checked by: is its claimed contextual state genuinely outside $\overline{\mathrm{conv}}(S_{\mathrm{df}})$, and is the obstruction genuinely absent from every finite sub-OML?

References

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